1. Create a Storage Account in Azure for Terraform State

```
askaniusz@askaniusz-Crosshair-16-HX-D14VFKG:~/Desktop/m07_sparksql_python_azure-master$ az group create --name [redacted] --location eastus
{
  "id": "/subscriptions[redacted]/resourceGroups/[redacted]",
  "location": "eastus",
  "managedBy": null,
  "name": "[redacted]",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
```

```
  },
  "keyPolicy": null,
  "kind": "StorageV2",
  "largeFileSharesState": null,
  "lastGeoFailoverTime": null,
  "location": "eastus",
  "minimumTlsVersion": "TLS1_0",
  "name": "[redacted]",
  "networkRuleSet": {
    "bypass": "AzureServices",
    "defaultAction": "Allow",
    "ipRules": [],
    "ipv6Rules": [],
    "resourceAccessRules": null,
    "virtualNetworkRules": []
  },
  "placement": null,
  "primaryEndpoints": {
    "blob": "https://[redacted]blob.core.windows.net/",
    "dfs": "https://s[redacted]dfs.core.windows.net/",
    "file": "https://s[redacted].file.core.windows.net/",
    "internetEndpoints": null,
    "ipv6Endpoints": null,
    "microsoftEndpoints": null,
    "queue": "https://[redacted]queue.core.windows.net/",
    "table": "https://s[redacted]s.table.core.windows.net/",
    "web": "https:/[redacted]z13.web.core.windows.net/"
  },
```

```
askaniusz@askaniusz-Crosshair-16-HX-D14VFKG:~/Desktop/m07_sparksql_python_azure-master$ az storage container
 create --name tfstate --account-name [redacted]

There are no credentials provided in your command and environment, we will query for account key for your st
orage account.
It is recommended to provide --connection-string, --account-key or --sas-token in your command as credential
s.

You also can add `--auth-mode login` in your command to use Azure Active Directory (Azure AD) for authorizat
ion if your login account is assigned required RBAC roles.
For more information about RBAC roles in storage, visit https://learn.microsoft.com/azure/storage/common/sto
rage-auth-aad-rbac-cli.

In addition, setting the corresponding environment variables can avoid inputting credentials in your command
. Please use --help to get more information about environment variable usage.
{
  "created": true
}
```

3. Update Terraform Configuration

main.tf

terraform > main.tf

```
# Setup azurerm as a state backend
terraform {
  backend "azurerm" {
    resource_group_name  = [redacted]"
    storage_account_name = "[redacted]" # Provide Storage Account name, where Terraform Remote state is stored
    container_name       = "[redacted]"
    key                  = [redacted]
  }
}

# Configure the Microsoft Azure Provider
provider "azurerm" {
  features {}
  subscription_id = [redacted]
}
```

4. Deploy Infrastructure with Terraform

```
askaniusz@askaniusz-Crosshair-16-HX-D14VFKG:~/Desktop/m07_sparksql_python_azure-master/terraform$ terrafo
output resource_group_name
"rg-d[redacted]pv"
askaniusz@askaniusz-Crosshair-16-HX-D14VFKG:~/Desktop/m07_sparksql_python_azure-master/terraform$
```

Name ↑	Subscription	Location
databricks-rg-rg-de	Azure subscription 1	West Europe
Network\	Azure subscription 1	West Europe
rg-dat	Azure subscription 1	East US
rg-de	Azure subscription 1	West Europe

Home › Resource Manager | Resource groups › Containers

data
Container

Successfully uploaded blob(s)
Successfully uploaded 973 blob(s).

Search

- Overview
- Diagnose and solve problems
- Access Control (IAM)
- Settings
  - Shared access tokens
  - Manage ACL
  - Access policy
  - Properties
  - Metadata

Add Directory | Upload | Refresh | Delete | Copy | Paste | Rename | Acquire lease

data

Authentication method: Access key (Switch to Microsoft Entra user account)

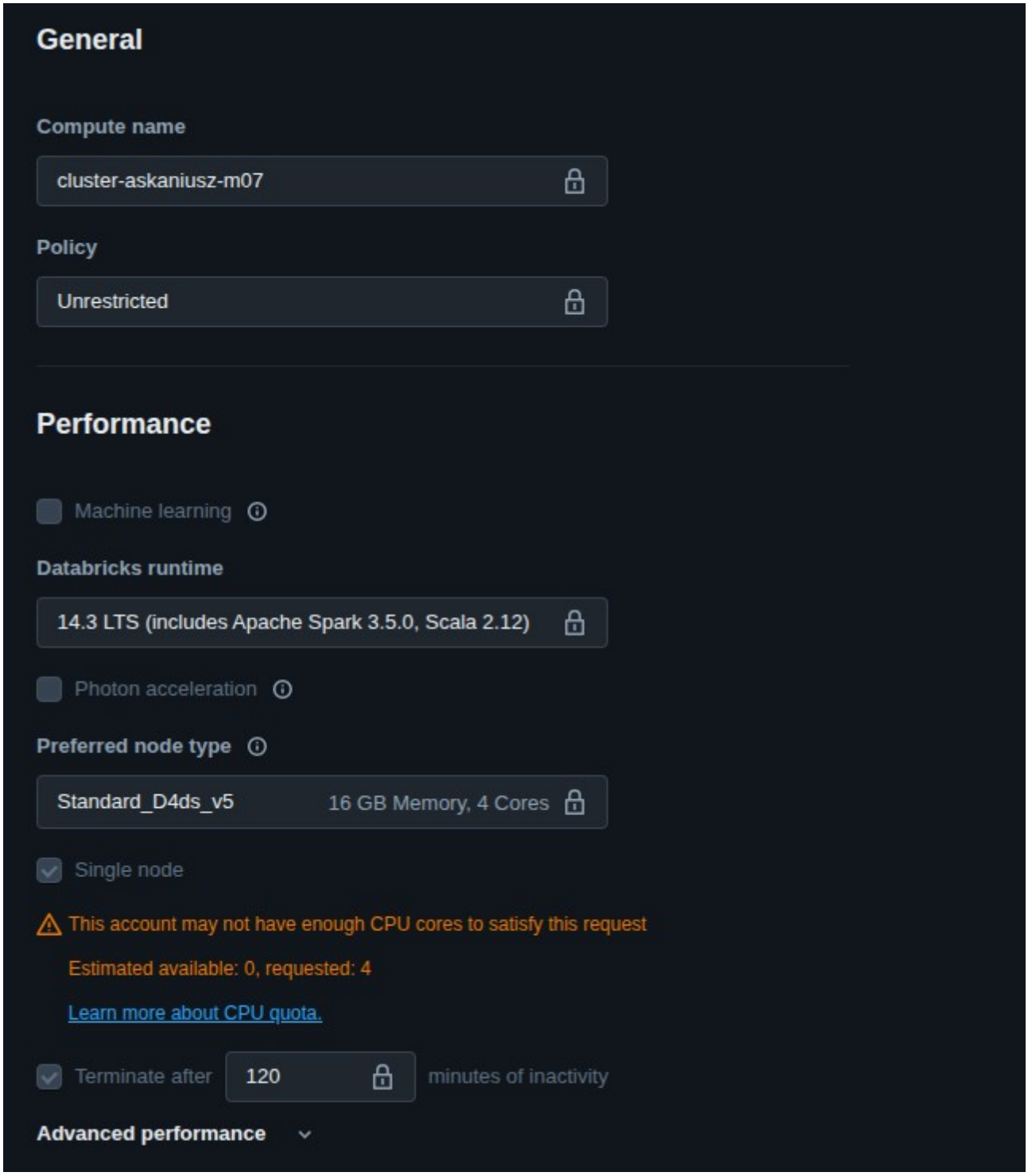
Search blobs by prefix (case-sensitive) | Only show active objects

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
expedia	2/21/2026, 10:40:17 AM				
hotel-weat...	2/21/2026, 10:44:11 AM				

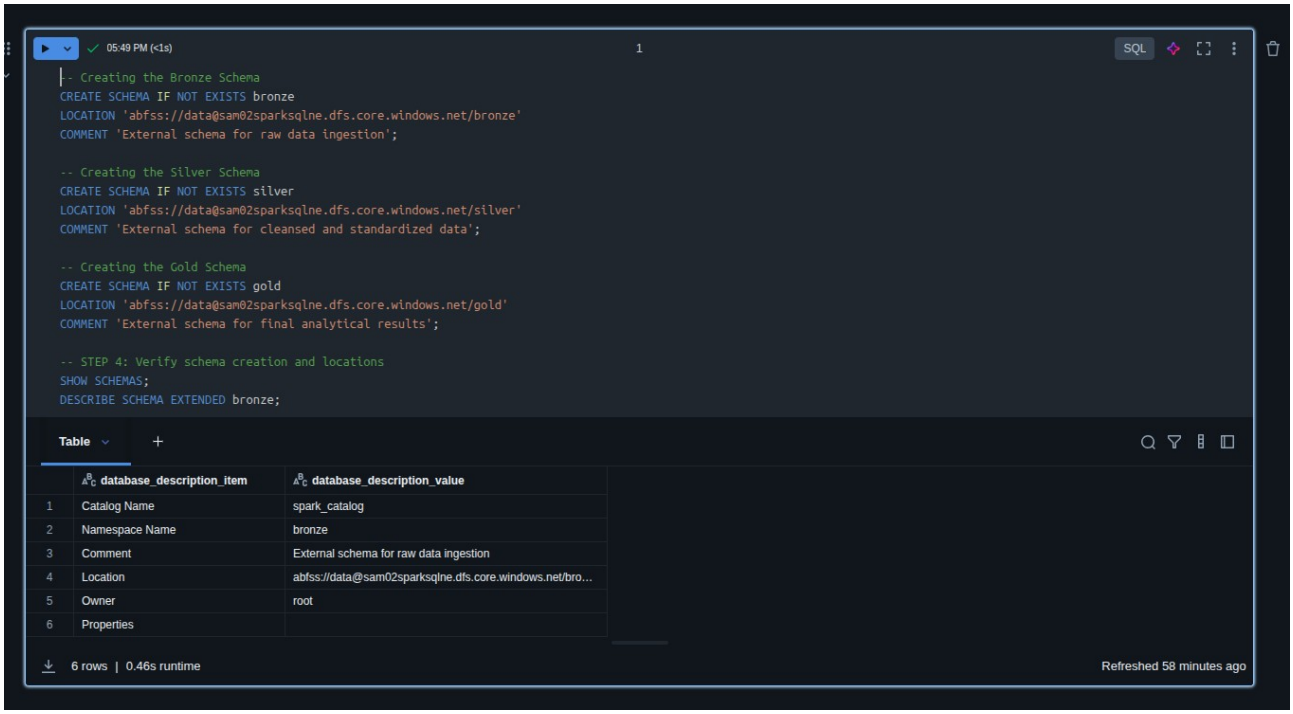
6. Launch notebooks on Databricks cluster

Create cluster



Create Databricks Notebook → 01_create_metadata

Create three schemas: bronze, silver and gold.



Encrypt Personally Identifiable Information (PII) fields to protect sensitive data.

06:31 PM (1s) 1 SQL

```sql
CREATE OR REPLACE FUNCTION bronze.encrypt_pii(col_value STRING)
RETURNS STRING
RETURN base64(aes_encrypt(col_value, '1234567890123456', 'GCM'));

CREATE OR REPLACE FUNCTION bronze.decrypt_pii(encrypted_value STRING)
RETURNS STRING
RETURN CAST(aes_decrypt(unbase64(encrypted_value), '1234567890123456', 'GCM') AS STRING);
```

OK

Load bronze data

```
06:32 PM (1s)                                                    1                                                    SQL
%run ./pii_utils
OK
+ Code   + Text   Assistant
```

```
06:11 PM (2m)                                                    2                                                    SQL
CREATE OR REPLACE TABLE bronze.expedia_raw
USING DELTA
LOCATION 'abfss://data@sam02sparksqlne.dfs.core.windows.net/bronze/expedia_raw'
AS
SELECT
    * EXCEPT (user_id, user_location_city),
    -- Re-adding columns as encrypted strings
    bronze.encrypt_pii(CAST(user_id AS STRING)) AS user_id,
    bronze.encrypt_pii(user_location_city) AS user_location_city
FROM read_files(
  'abfss://data@sam02sparksqlne.dfs.core.windows.net/expedia/',
  format => 'avro'
);

-- 2. Ingest Hotel Weather Data (Parquet format)
-- Note: The path is 'hotel-weather' as verified by your file listing
CREATE OR REPLACE TABLE bronze.hotel_weather_raw
USING DELTA
LOCATION 'abfss://data@sam02sparksqlne.dfs.core.windows.net/bronze/hotel_weather_raw'
AS
SELECT * FROM read_files(
  'abfss://data@sam02sparksqlne.dfs.core.windows.net/hotel-weather/',
  format => 'parquet'
);

-- Documentation
COMMENT ON TABLE bronze.expedia_raw IS 'Raw Expedia bookings with encrypted PII';
COMMENT ON TABLE bronze.hotel_weather_raw IS 'Raw Weather data per hotel and date';
(22) Spark Jobs
OK
```

Bronze to Silver

06:32 PM (1s) 1 SQL

```
%run ./pii_utils
```

OK

+ Code + Text Assistant

06:17 PM (1m) 2 SQL

```sql
-- Creating Silver table for Expedia
CREATE OR REPLACE TABLE silver.expedia_processed
USING DELTA
LOCATION 'abfss://data@sam02sparksqlne.dfs.core.windows.net/silver/expedia_processed'
AS
WITH decrypted_data AS (
    -- Step 1: Decrypt PII fields for processing
    SELECT
        * EXCEPT (user_id, user_location_city),
        bronze.decrypt_pii(user_id) AS user_id_raw,
        bronze.decrypt_pii(user_location_city) AS city_raw
    FROM bronze.expedia_raw
),
cleaned_data AS (
    -- Step 2: Apply cleaning and standardization
    SELECT DISTINCT
        CAST(hotel_id AS BIGINT) AS hotel_id,
        TRIM(user_id_raw) AS user_id_clean,
        -- Standardizing check-in/out date formats
        CAST(srch_ci AS TIMESTAMP) AS srch_ci,
        CAST(srch_co AS TIMESTAMP) AS srch_co,
        -- Handling empty strings and trimming
        NULLIF(TRIM(city_raw), '') AS city_clean
    FROM decrypted_data
    -- Filter out records with nulls in critical fields
    WHERE hotel_id IS NOT NULL
      AND srch_ci IS NOT NULL
      AND srch_co IS NOT NULL
)
-- Step 3: Re-encrypt and store
SELECT
    * EXCEPT (user_id_clean, city_clean),
    bronze.encrypt_pii(user_id_clean) AS user_id,
    bronze.encrypt_pii(city_clean) AS user_location_city
FROM cleaned_data;

COMMENT ON TABLE silver.expedia_processed IS 'Standardized Expedia data with re-encrypted PII';
```

(12) Spark Jobs

OK

06:23 PM (7s) 3

```sql
-- Creating Silver table for Weather with corrected column mapping
-- Using avg_tmpr_c as the temperature source based on schema suggestions
CREATE OR REPLACE TABLE silver.hotel_weather_processed
USING DELTA
LOCATION 'abfss://data@sam02sparksqlne.dfs.core.windows.net/silver/hotel_weather_processed'
AS
SELECT DISTINCT
    -- Mapping 'id' from Bronze to 'hotel_id' in Silver for consistency
    CAST(id AS BIGINT) AS hotel_id,
    -- Constructing a proper date from year, month, and day columns
    CAST(make_date(year, month, day) AS DATE) AS weather_date,
    -- Using the Celsius column as identified in the schema error
    CAST(avg_tmpr_c AS DOUBLE) AS avg_temperature
FROM bronze.hotel_weather_raw
WHERE id IS NOT NULL
  AND year IS NOT NULL
  AND month IS NOT NULL
  AND day IS NOT NULL;

COMMENT ON TABLE silver.hotel_weather_processed IS 'Cleaned weather data with temperature in Celsius and mapped hotel_id';
```

(12) Spark Jobs

OK

Silver to Gold

```
%run ./pii_utils
```

OK

```
-- Top 10 Hotels with Maximum Absolute Temperature Difference by Month
CREATE OR REPLACE TABLE gold.top_10_temp_diff_monthly
AS
SELECT
    hotel_id,
    MONTH(weather_date) AS month,
    (MAX(avg_temperature) - MIN(avg_temperature)) AS abs_temp_diff
FROM silver.hotel_weather_processed
GROUP BY hotel_id, month
ORDER BY ABS(abs_temp_diff) DESC
LIMIT 10;
```

(8) Spark Jobs

Table

num_affected_rows	num_inserted_rows

No rows returned

0 rows | 6.04s runtime

Refreshed 17 minutes ago

```
-- Top 10 busiest hotels per month based on visit counts
CREATE OR REPLACE TABLE gold.top_10_busiest_hotels_monthly
AS
SELECT
    hotel_id,
    MONTH(srch_ci) AS month,
    COUNT(*) AS visit_count
FROM silver.expedia_processed
GROUP BY hotel_id, month
ORDER BY visit_count DESC
LIMIT 10;
```

(8) Spark Jobs

Table

num_affected_rows	num_inserted_rows

```
-- Temperature trend (end vs start) for visits longer than 7 days
CREATE OR REPLACE TABLE gold.weather_trend_extended_stay
AS
WITH extended_stays AS (
    SELECT hotel_id, srch_ci, srch_co
    FROM silver.expedia_processed
    WHERE DATEDIFF(srch_co, srch_ci) > 7
)
SELECT
    es.hotel_id,
    es.srch_ci,
    es.srch_co,
    (w_end.avg_temperature - w_start.avg_temperature) AS temp_trend,
    (SELECT AVG(avg_temperature)
     FROM silver.hotel_weather_processed
     WHERE hotel_id = es.hotel_id
       AND weather_date BETWEEN CAST(es.srch_ci AS DATE) AND CAST(es.srch_co AS DATE)) AS avg_stay_temp
FROM extended_stays es
JOIN silver.hotel_weather_processed w_start
  ON es.hotel_id = w_start.hotel_id AND CAST(es.srch_ci AS DATE) = w_start.weather_date
JOIN silver.hotel_weather_processed w_end
  ON es.hotel_id = w_end.hotel_id AND CAST(es.srch_co AS DATE) = w_end.weather_date;
```

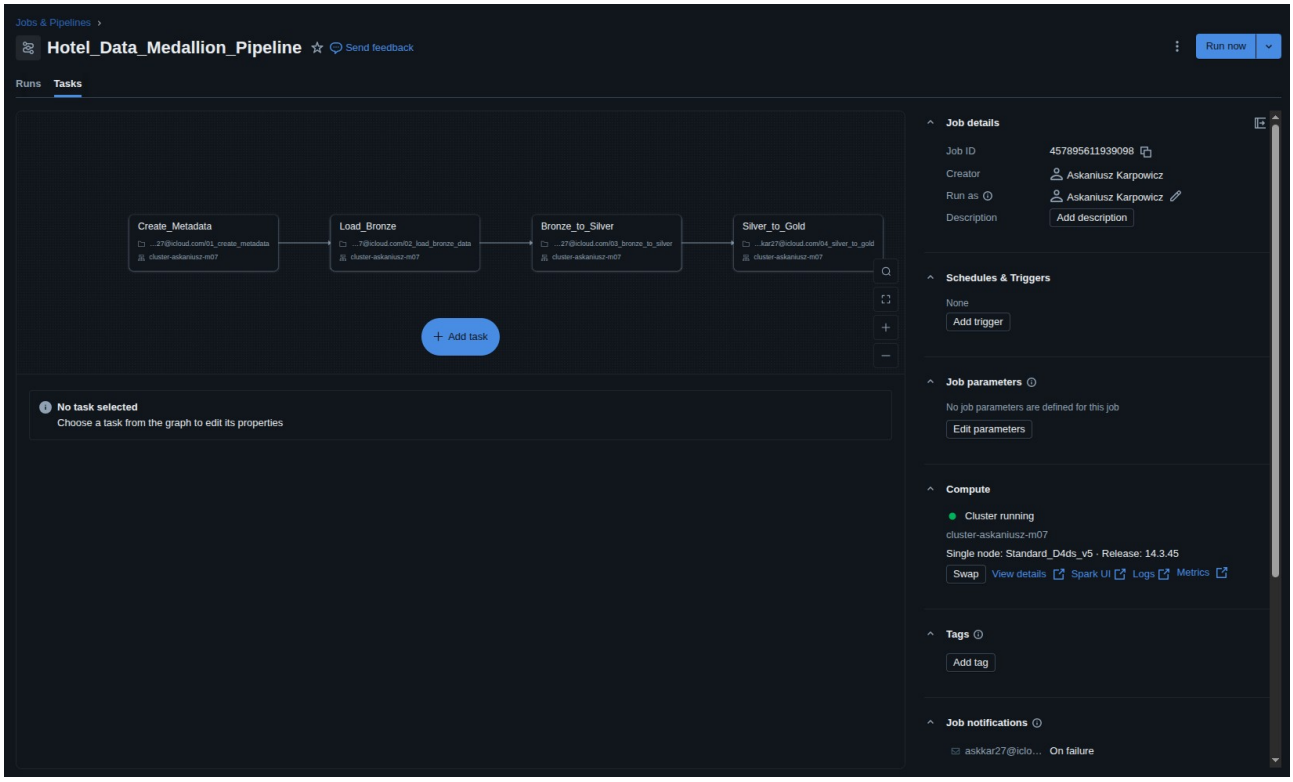
(11) Spark Jobs

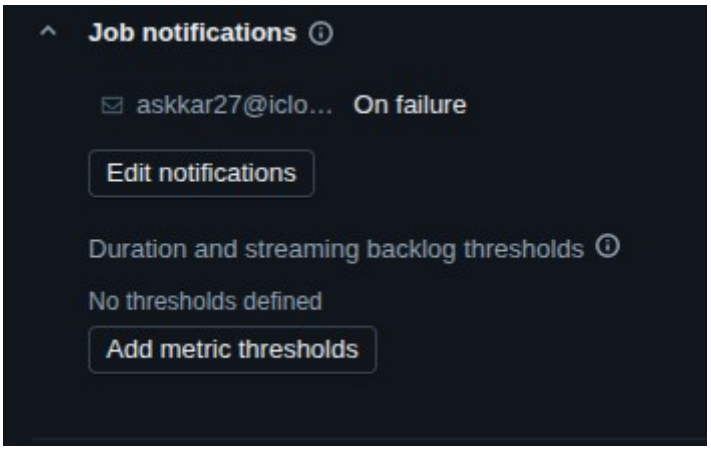
Table

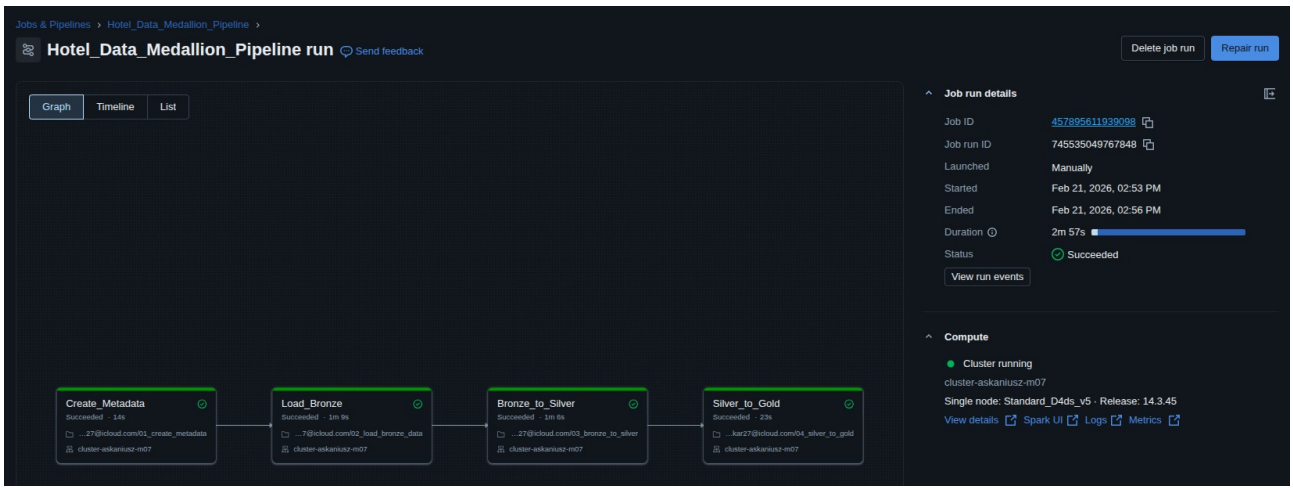
num_affected_rows	num_inserted_rows

No rows returned

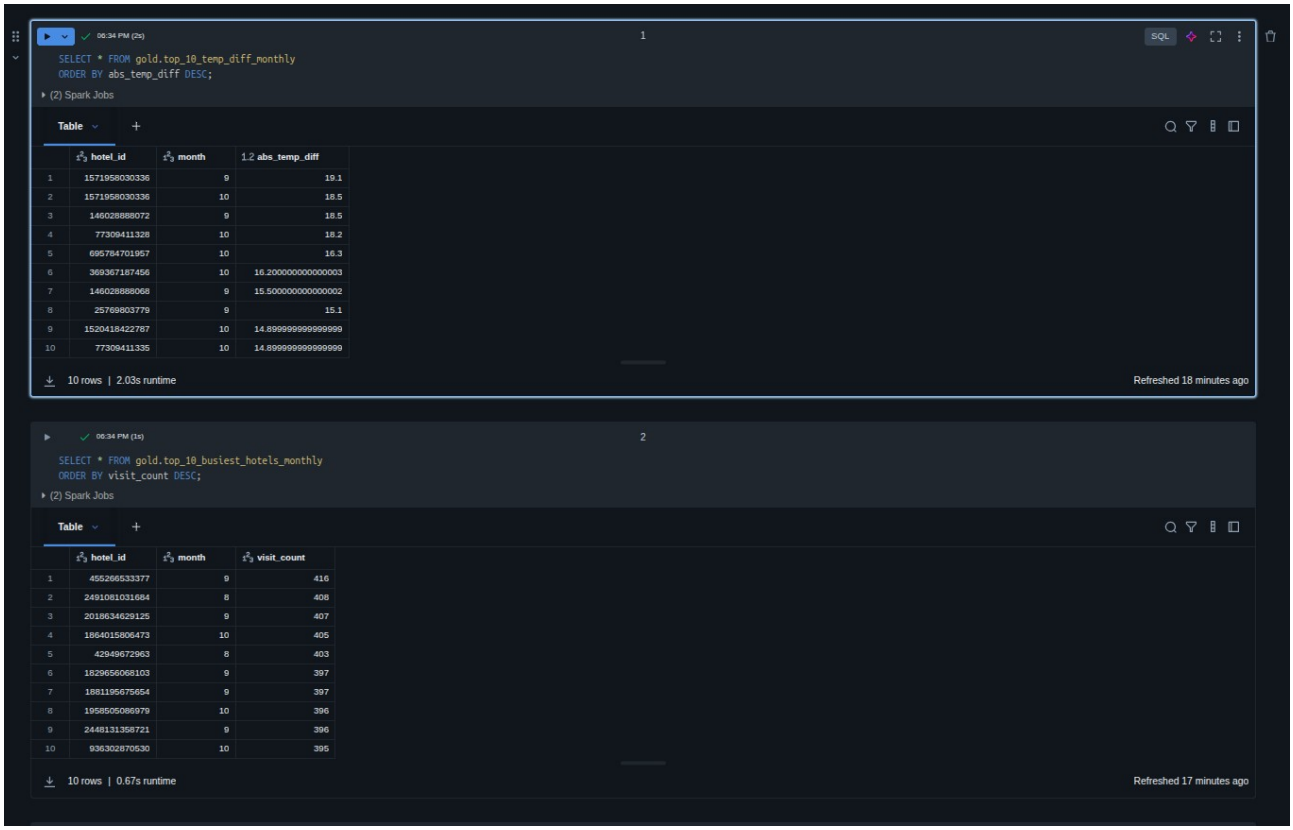
Orchestrate the Jobs via Databricks Workflow






Jobs & Pipelines › Hotel_Data_Medallion_Pipeline ›
Hotel_Data_Medallion_Pipeline run
Send feedback
Delete job run
Repair run
Graph
Timeline
List
Create_Metadata
Succeeded · 14s
...27@icloud.com/01_create_metadata
cluster-askaniusz-m07
Load_Bronze
Succeeded · 1m 9s
...7@icloud.com/02_load_bronze_data
cluster-askaniusz-m07
Bronze_to_Silver
Succeeded · 1m 6s
...27@icloud.com/03_bronze_to_silver
cluster-askaniusz-m07
Silver_to_Gold
Succeeded · 23s
...kar27@icloud.com/04_silver_to_gold
cluster-askaniusz-m07
Job run details
Job ID
457895611939098
Job run ID
745535049767848
Launched
Manually
Started
Feb 21, 2026, 02:53 PM
Ended
Feb 21, 2026, 02:56 PM
Duration
2m 57s
Status
Succeeded
View run events
Compute
Cluster running
cluster-askaniusz-m07
Single node: Standard_D4ds_v5 · Release: 14.3.45
View details
Spark UI
Logs
Metrics

Check data after Workflow



```
SELECT * FROM gold.top_10_temp_diff_monthly
ORDER BY abs_temp_diff DESC;
```

(2) Spark Jobs

Table

	hotel_id	month	abs_temp_diff
1	1571958030336	9	19.1
2	1571958030336	10	18.5
3	146028888072	9	18.5
4	77309411328	10	18.2
5	695784701957	10	16.3
6	369367187456	10	16.200000000000003
7	146028888068	9	15.500000000000002
8	25769803779	9	15.1
9	1520418422787	10	14.899999999999999
10	77309411335	10	14.899999999999999

10 rows | 2.03s runtime

Refreshed 18 minutes ago

```
SELECT * FROM gold.top_10_busiest_hotels_monthly
ORDER BY visit_count DESC;
```

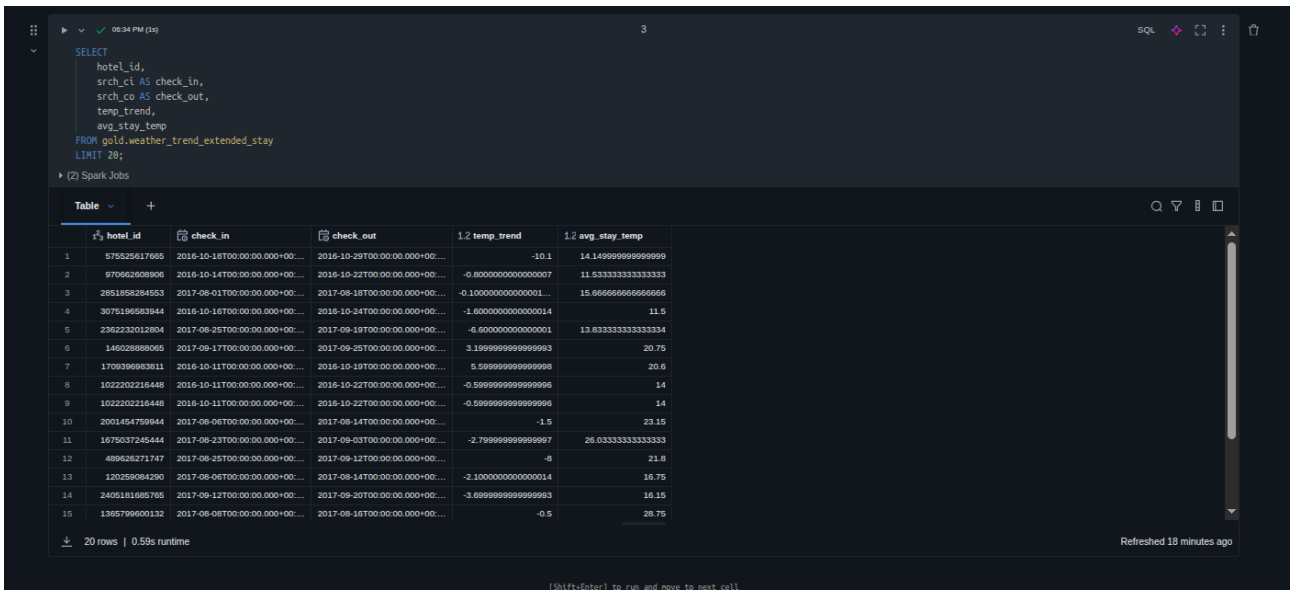
(2) Spark Jobs

Table

	hotel_id	month	visit_count
1	455266533377	9	416
2	2491081031684	8	408
3	2018634629125	9	407
4	1864015806473	10	405
5	42949672963	8	403
6	1829656068103	9	397
7	1881195675654	9	397
8	1958505086979	10	396
9	2448131358721	9	396
10	936302870530	10	395

10 rows | 0.67s runtime

Refreshed 17 minutes ago



```
SELECT
    hotel_id,
    srch_ci AS check_in,
    srch_co AS check_out,
    temp_trend,
    avg_stay_temp
FROM gold.weather_trend_extended_stay
LIMIT 20;
```

(2) Spark Jobs

Table

	hotel_id	check_in	check_out	temp_trend	avg_stay_temp
1	575525617665	2016-10-18T00:00:00.000+00:...	2016-10-29T00:00:00.000+00:...	-10.1	14.149999999999999
2	970662608906	2016-10-14T00:00:00.000+00:...	2016-10-22T00:00:00.000+00:...	-0.8000000000000007	11.533333333333333
3	2851858284553	2017-08-01T00:00:00.000+00:...	2017-08-18T00:00:00.000+00:...	-0.100000000000001...	15.666666666666666
4	3075196583944	2016-10-16T00:00:00.000+00:...	2016-10-24T00:00:00.000+00:...	-1.6000000000000014	11.5
5	2362232012804	2017-08-25T00:00:00.000+00:...	2017-09-19T00:00:00.000+00:...	-6.600000000000001	13.833333333333334
6	146028888065	2017-09-17T00:00:00.000+00:...	2017-09-25T00:00:00.000+00:...	3.1999999999999993	20.75
7	1709396963811	2016-10-11T00:00:00.000+00:...	2016-10-19T00:00:00.000+00:...	5.599999999999998	20.6
8	1022202216448	2016-10-11T00:00:00.000+00:...	2016-10-22T00:00:00.000+00:...	-0.5999999999999996	14
9	1022202216448	2016-10-11T00:00:00.000+00:...	2016-10-22T00:00:00.000+00:...	-0.5999999999999996	14
10	2001454759944	2017-08-06T00:00:00.000+00:...	2017-08-14T00:00:00.000+00:...	-1.5	23.15
11	1675037245444	2017-08-23T00:00:00.000+00:...	2017-09-03T00:00:00.000+00:...	-2.799999999999997	26.033333333333333
12	489626271747	2017-08-25T00:00:00.000+00:...	2017-09-12T00:00:00.000+00:...	-8	21.8
13	120259084290	2017-08-06T00:00:00.000+00:...	2017-08-14T00:00:00.000+00:...	-2.1000000000000014	16.75
14	2405181685765	2017-09-12T00:00:00.000+00:...	2017-09-20T00:00:00.000+00:...	-3.6999999999999993	16.15
15	1365799600132	2017-08-08T00:00:00.000+00:...	2017-08-16T00:00:00.000+00:...	-0.5	28.75

20 rows | 0.59s runtime

Refreshed 18 minutes ago

7. Destroy Infrastructure

```
azurerm_resource_group.bdcc: Destruction complete after 47s
random_string.suffix: Destroying... [id=xdpv]
random_string.suffix: Destruction complete after 0s
Releasing state lock. This may take a few moments...

Destroy complete! Resources: 5 destroyed.
askaniusz@askaniusz-Crosshair-16-HX-D14VFKG:~/Desktop/m07_sparksql_python_azure-master/terraform$
```